

# Szymon Sarnowicz

[szymon@sarnowicz.net](mailto:szymon@sarnowicz.net) | 425-802-6406 | [linkedin.com/in/szymonsarnowicz](https://linkedin.com/in/szymonsarnowicz) | [github.com/zymoncone](https://github.com/zymoncone) | [szymon.wiki](https://szymon.wiki)

## Experience

### Readiscover.app – Founding Technical Lead | Jan 2026 – present

- Built full-stack semantic search platform for long-form text retrieval. Engineered dynamic chunking strategy (1,200-char chunks, 3-sentence overlap) and deployed multi-layer caching architecture: GCS for persistent storage, Redis for sub-second cache access (24-hour TTL).
- Implemented LLM query reformulation using Gemini 2.5 Flash (temperature 0.0 for deterministic output) before embedding with Google text-embedding-005 model. System design prioritizes trustworthiness: returns only continuous verbatim passages, eliminating hallucination risk (vs. 43.6% for GPT-5.1 baseline).
- Conducted rigorous ablation testing across chunking parameters; evaluated against GPT-5.1 on 50 test questions. Results: 34% top-3 accuracy with zero hallucinations, superior proximity to correct passages. Capstone project published with co-authors at the University of Michigan.

### General Motors – Software Engineer II | Feb 2023 – present

- Owned end-to-end redesign of safety-critical fault detection system in C/C++, implementing sensor data processing, signal filtering algorithms, and state machine logic for fault progression. Architected persistent state management to prevent critical system failures, helping avoid ~\$480M in potential recalls.
- Architected and deployed production-level diagnostic system from concept to deployment; designed state monitoring logic, implemented condition detection algorithms, and integrated alert routing protocols across distributed services. Increased system reliability by refactoring fault-response behavior across microservices and developing a Python testing framework for safety-critical validation.

### Data4Good – University of Michigan – Lead Developer, Hangul Project | Dec 2023 – Dec 2024 (Volunteer)

- Developed Svelte web application for United Nations (UN) document summarization using Apache Tika for HTML parsing and BART model for NLP-based sentence extraction. Presented product to UN representatives for feedback and validation. Solved resource constraints by redesigning API pipeline into multiple stages with exponential backoff, increasing success rate from 53% to 92%. Demonstrated systems-thinking approach to handling bottlenecks in constrained environments.

### General Motors – Program Manager, Autonomous Systems (Ultra Cruise) | Jul 2021 – Feb 2023

- Tripled data collected per sprint by automating data-request workflows, improving visibility for perception and autonomy engineering teams. Enabled faster iteration cycles for sensor fusion and real-time decision-making pipelines in autonomous vehicle development.
- Authored comprehensive troubleshooting guides and diagnostics workflows that enabled cross-functional teams to independently resolve camera, radar, and LiDAR system issues. Bridged hardware integration and software systems, improving sensor calibration diagnostics and reducing development cycle time.

### General Motors – Software Engineer (Rotational Program) | Sep 2019 – Jul 2021

- Automated calibration data management processes across embedded systems, eliminating 1 hour of manual data entry work per day. Gained foundational experience in embedded systems, real-time data handling, sensor calibration, and automotive software development practices.

## Education

**Master's in Applied Data Science**, University of Michigan, summa cum laude | Dec 2025

**B.S. Mechanical Engineering**, University of Washington, cum laude | Jun 2019

## Technical Skills & Awards

**Languages:** Python, C, C++, JavaScript

**Systems:** C++ for embedded systems, real-time signal processing, state machines

**ML/AI:** LLMs, semantic search, embeddings, NLP, RAG

**Cloud & Infrastructure:** GCP (Vertex AI, Cloud Run), AWS (Lambda, DynamoDB), Docker, Redis

**Tools:** React, FastAPI, Git, Linux

**Best Cybersecurity Hack**, MHack16 (350+ participants) | Dec 2023